

Roll No.

--	--	--	--	--	--	--	--

MID SEMESTER RE EXAMINATION 2024-25

Name of the Course: B.Tech. PE

Semeste VI

Name of the Paper: Data SciencePaper Code: TPE 602

Time: 1:30 Hrs

Maximum Marks: 50

Note: -

- (i) All questions are compulsory.
- (ii) Do any one of the (a) or (b) in each question
- (iii) Each question carries 10 marks

Q1	(10marks)	CO1																														
(a)	Define Analytics and Data Science. Explain how analytics is applied in various industries and how data science differs from traditional data analysis. Provide real-world examples of its applications.																															
	OR																															
(b)	Explain the Analytics Methodology and Problem-Solving Frameworks. How do businesses use analytical models to solve complex problems? Discuss key problem-solving techniques with suitable examples.																															
Q2	(10marks)	CO1, CO2																														
(a)	What are Models and Algorithms in Data Science? Explain different types of models used in analytics and how algorithms play a role in predictive analytics.																															
	OR																															
(b)	Discuss the importance of Machine Learning in Data Science. How do supervised and unsupervised learning techniques contribute to model building? Provide examples of each.																															
Q3	(10marks)	CO1, CO3																														
(a)	Explain the SQL SELECT statement and its role in database querying. Use the following table <i>Employees</i> to demonstrate how to retrieve specific columns and apply filters using the WHERE clause.																															
<table><tr><th>Emp_ID</th><th>Name</th><th>Age</th><th>Department</th><th>Salary</th></tr><tr><td>101</td><td>Alice</td><td>30</td><td>IT</td><td>60000</td></tr><tr><td>102</td><td>Bob</td><td>25</td><td>HR</td><td>50000</td></tr><tr><td>103</td><td>Charlie</td><td>28</td><td>IT</td><td>55000</td></tr><tr><td>104</td><td>David</td><td>35</td><td>Finance</td><td>70000</td></tr><tr><td>105</td><td>Eva</td><td>27</td><td>HR</td><td>52000</td></tr></table>			Emp_ID	Name	Age	Department	Salary	101	Alice	30	IT	60000	102	Bob	25	HR	50000	103	Charlie	28	IT	55000	104	David	35	Finance	70000	105	Eva	27	HR	52000
Emp_ID	Name	Age	Department	Salary																												
101	Alice	30	IT	60000																												
102	Bob	25	HR	50000																												
103	Charlie	28	IT	55000																												
104	David	35	Finance	70000																												
105	Eva	27	HR	52000																												
Write queries to:																																
- Select all employees from the IT department. - Retrieve names and salaries of employees earning more than 55,000. - Find employees whose names start with 'A'.																																

	OR
(b)	How does the ORDER BY clause help in sorting data? Explain its significance in database management using the Employees table. Write SQL queries to: 1. Display all employees sorted by age in ascending order. 2. List employees sorted by salary in descending order. 3. Retrieve employees from the HR department sorted alphabetically by name.

Q4 (10 marks)

CO1, CO5

- (a) What are SQL Joins? Explain different types of joins (INNER JOIN, LEFT JOIN, RIGHT JOIN, FULL JOIN) using the Employees table and the Departments table.

Dept_ID	Dept_Name
1	IT
2	HR
3	Finance
4	Marketing

Write queries to:

1. Display all employees along with their department names using INNER JOIN.
2. Show all employees along with department names, ensuring all employees are displayed even if they do not have a department assigned.

	OR
(b)	Discuss the concept of Aggregate Functions in SQL. Using the Employees table, write queries to: 1. Find the total salary paid to all employees. 2. Calculate the average salary of employees in the IT department. 3. Count the number of employees in each department.

Q5 (10 marks)

CO1, CO5

- (a) What is the GROUP BY clause in SQL? How is it used for grouping data? Using the Employees table, write SQL queries to:
1. Group employees by department and calculate the total salary for each department.
 2. Count the number of employees in each department.
 3. Find the highest salary in each department.

OR

(b) Explain the HAVING clause in SQL. How is it different from the WHERE clause? Using the Employees table, write queries to:	
---	--

1. Display departments where the total salary exceeds 100,000.
2. Show departments with more than 2 employees.
3. Find departments where the average salary is greater than 60,000.

